

Research projects 2024-2025

Computational chemistry and modern machine learning techniques for the selection of short amphiphilic peptide sequences and the conception of new theranostic nanovectors: from structure to self-assembling properties

Supervisor:

Name: Fanny d'Orlyé

Phone: +33 (0)1 85 78 41 58

E-mail: fanny.dorlye@chimieparistech.psl.eu

Host Laboratory:

Affiliation: Chimie ParisTech/CNRS

Lab Name : i-CLeHS, UMR 8060 CNRS, SEISAD Team

Address : 11, rue Pierre et Marie Curie, 75005 Paris

Partners or collaborations :

Name: Avagliano Davide

Phone : +33 (0)1 85 78 41 72

E-mail: davide.avagliano@chimieparistech.psl.eu

Affiliation: Chimie ParisTech/CNRS

Lab Name : i-CLeHS, UMR 8060 CNRS, CTM Team

Address : 11, rue Pierre et Marie Curie, 75005 Paris

Project description :

The development of new alternative therapeutic strategies for the treatment of cancer while limiting undesirable side effects of conventional chemotherapy is essential. Nanomedicine is an effective approach to meet this requirement, as the encapsulation of antitumor molecules in biocompatible nanovectors should make it possible to reduce their toxicity while improving their circulation and targeting *in vivo*¹.

In this context, the SEISAD team has expertise in the design of new nanovectors based on short peptide self-assembly in monodisperse and size-controlled cavitory supramolecular nanostructures. Indeed, such supramolecular architectures offer the possibility of encapsulating small molecules for better preservation of their activity but also better control of their release in response to different stimuli (pH, thermal, photonic, enzymatic ...) ² They also feature surface chemical reactivity via amino- and carboxyl groups, enabling chemical grafting of targeting ligands and imaging probes (optical, MRI, PET), in addition to their intrinsic biocompatibility and biodegradability.

¹ Signal Transduct. Target. Ther., 3 (2018) 7

² Current Opinion in Colloid & Interface Science 45 (2020) 1–13

So far, we have studied a peptide sequence with an amphiphilic character including an RGD motif for active targeting of $\alpha v \beta 3$ integrin receptors overexpressed in cancer cells, and able to self-assemble into pH-responsive micelle-like nanostructures³. A continuous-flow solid-phase peptide synthesis (SPPS) was developed and optimized in the laboratory through an iterative process based on CE-ESI-MS analysis of the peptide product. The exact sequence was then functionalized with a gadolinium complex and the nanovector successfully formulated based on the co-assembly of the peptide and its conjugate in different proportions. Nonetheless, spontaneously organized nano-rods could only be characterized (DLS, zetametry, TDA, TEM and CE-UV) in strongly basic media (pH < 11), without confirmation of the stoichiometry of assemblies⁴.

The combination of traditional computational chemistry and modern machine learning techniques has already demonstrated its potential to predict the properties/reactivity of short synthetic peptides and their derivatives⁵ but also of peptide self-assemblies⁶. The scope of this internship will be to benefit from the first set of experimental data acquired in the SEISAD team together with already published data on self-assembling short amphiphilic peptides to build high-performance machine learning models, enriched with knowledge from the field of theoretical and computational chemistry, to predict self-assembly⁷. The aim will be to rapidly identify one or more sequences likely to lead to the right nano-vector characteristics: cavitary nano-architectures thermodynamically stable in the pH 6-7.4 range. Complementary, molecular dynamics will be performed to assess (i) the thermodynamic stability of self-assembled structures resulting from selected sequences in physiological environments and over time, and (ii) the impact of grafting imaging probes on self-assembly properties of these peptides. Eventually, the self-assembly will be evaluated experimentally in the lab.

The intern will have the excellent opportunity to integrate the Synthesis, Electrochemistry, Imaging and Analytical Systems for Diagnosis (SEISAD) team recognized for its expertise in the design and development of innovative tools for the early detection of pathological signals using chemical and analytical methodologies and work in collaboration with the Chemical Theory and Modelling (CTM) team recognized for its expertise in the development of new approaches for in-silico modeling of the reactivity and properties of molecules and extended systems in complex environments.

Keywords: peptide sequence selection; self-assembling peptide; nanostructured biomaterials; Computational chemistry; Machine learning.

³ J. Liang et al., *Colloids Surf. B* 114 (2014) 398–403

⁴ A. Am et al. *J. Chromatogr. A* 1713 (2024) 464496

⁵ J.A. Fuchs et al., *MedChemComm* 9 (2018) 1538-1546

⁶ H. Cox et al., *Biomacromolecules* 20 (2019) 1719-1730

⁷ N. Cao et al., *Nano Today* 55 (2024) 102160